

GLP LABORATÓRIOS

OIL & GAS · NORM

Solutions for the Oil & Gas sector

Radiometric analysis and NORM characterization for onshore, offshore and decommissioning operations.

- Committed to the future
- Brazilian laboratory
- Broadest scope of radiometric analyses

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02 · CONTEXT

NORM in the Oil & Gas industry

During oil and gas exploration, production and processing, naturally occurring radioactive materials can become concentrated in fluids, scale, sludge, sediments and operational waste.

These materials, known as **NORM** (Naturally Occurring Radioactive Material), may be present at different stages of the operation and require investigation, characterization and proper management.



Where can NORM be present?

01 Produced water	02 Well fluids
03 Sludge and sediments	04 Scale
05 Solid waste	06 Pipelines and piping
07 Tanks and separators	08 Process equipment
09 FPSOs and offshore units	10 Maintenance and cleaning materials
11 Decommissioning structures	



04 · APPLICATION

Offshore operations and FPSOs

On FPSOs, oil tankers and other offshore units, waste can accumulate in tanks, piping and equipment during operation.

During cleaning and maintenance activities, sludge, scale and other materials are removed and packaged for subsequent characterization.

Radiometric analysis makes it possible to identify the presence and concentration of natural radionuclides, providing technical information for the classification and management of these materials.

Why characterize NORM?

01

Occupational safety

Assessment of handling conditions and the radiation protection measures required.

02

Material classification

Determination of the activity concentration of the radionuclides present.

03

Waste management

Information for the storage, handling, treatment or disposal of materials.

04

Environmental protection

Technical data to support the proper management of potentially radioactive waste.

GLP solutions for O&G

- Radionuclide analysis in produced water
- NORM characterization
- Radiological analysis of solids and production waste
- Scale characterization
- Analysis of pipeline, tank and separator waste
- Analysis of materials from offshore operations
- Characterization of materials for handling or disposal
- Radiological characterization for decommissioning



07 · MATRIX

Produced water

Produced water is one of the main matrices of radiological interest in the oil and gas industry. GLP performs analyses for the identification and quantification of natural radionuclides, making it possible to assess the presence of radioactivity associated with the geological characteristics of the reservoir and production processes.

- Radiological monitoring
- Matrix characterization
- Process assessment
- Decision-making support

Solids, sludge and scale

Natural radionuclides can concentrate in different materials throughout the operation.

- Equipment and piping scale
- Tank sludge
- Sediments
- Separator waste
- Materials removed during cleaning and maintenance
- Production system waste

Analytical characterization makes it possible to determine the **activity concentration** present in these materials.

Pipelines, tanks and equipment

Maintenance, cleaning, replacement or equipment removal operations can generate waste or materials that need to be radiologically assessed. GLP performs analysis of samples from:

- **Pipelines**
- **Piping**
- **Tanks**
- **Separators**
- **Process equipment**
- **Structures and components**

10 · FINAL CYCLE

Decommissioning

The decommissioning of oil and gas facilities requires detailed knowledge of the conditions of the materials and structures involved. Radiological characterization can support the identification of materials containing NORM before processes of:

- 1 · Removal
- 2 · Transport
- 3 · Storage
- 4 · Reuse
- 5 · Treatment
- 6 · Final disposal

The analytical results provide a technical basis for planning the next stages of the process.



Analytical techniques

The choice of methodology depends on the matrix, the radionuclides of interest and the objectives of the investigation.

Alpha Spectrometry

Applied according to the radionuclides and objectives of the investigation.

Gamma Spectrometry

An important technique for the identification and quantification of gamma emitters.

Liquid Scintillation

Used in specific analyses depending on the matrix and radionuclide.

Proportional Counter

Gas-flow counter for specific applications.

Main radionuclides of interest

Depending on the matrix and application, the investigation may involve natural radionuclides associated with the uranium and thorium series, including:

Radium-226
226Ra

Radium-228
228Ra

Lead-210
210Pb

Uranium
Radioactive series

Thorium
Radioactive series

Other
Depending on the project

From collection to results

Five stages between identifying the need and the final technical result.

1

Identifying the need

Definition of the matrix and the objective of the characterization.

2

Sampling and packaging

Obtaining representative samples of the material.

3

Laboratory analysis

Application of the appropriate analytical methodology.

4

Quantification

Determination of the activity concentration of the radionuclides.

5

Technical result

Traceable data to support material management decisions.

Quality and reliability

A facility specialized in ionizing radiation testing and a quality system based on ABNT NBR ISO/IEC 17025. The combination of technical team, analytical infrastructure and specialized methodologies makes it possible to deliver reliable data and traceable results.

ABNT NBR ISO/IEC 17025

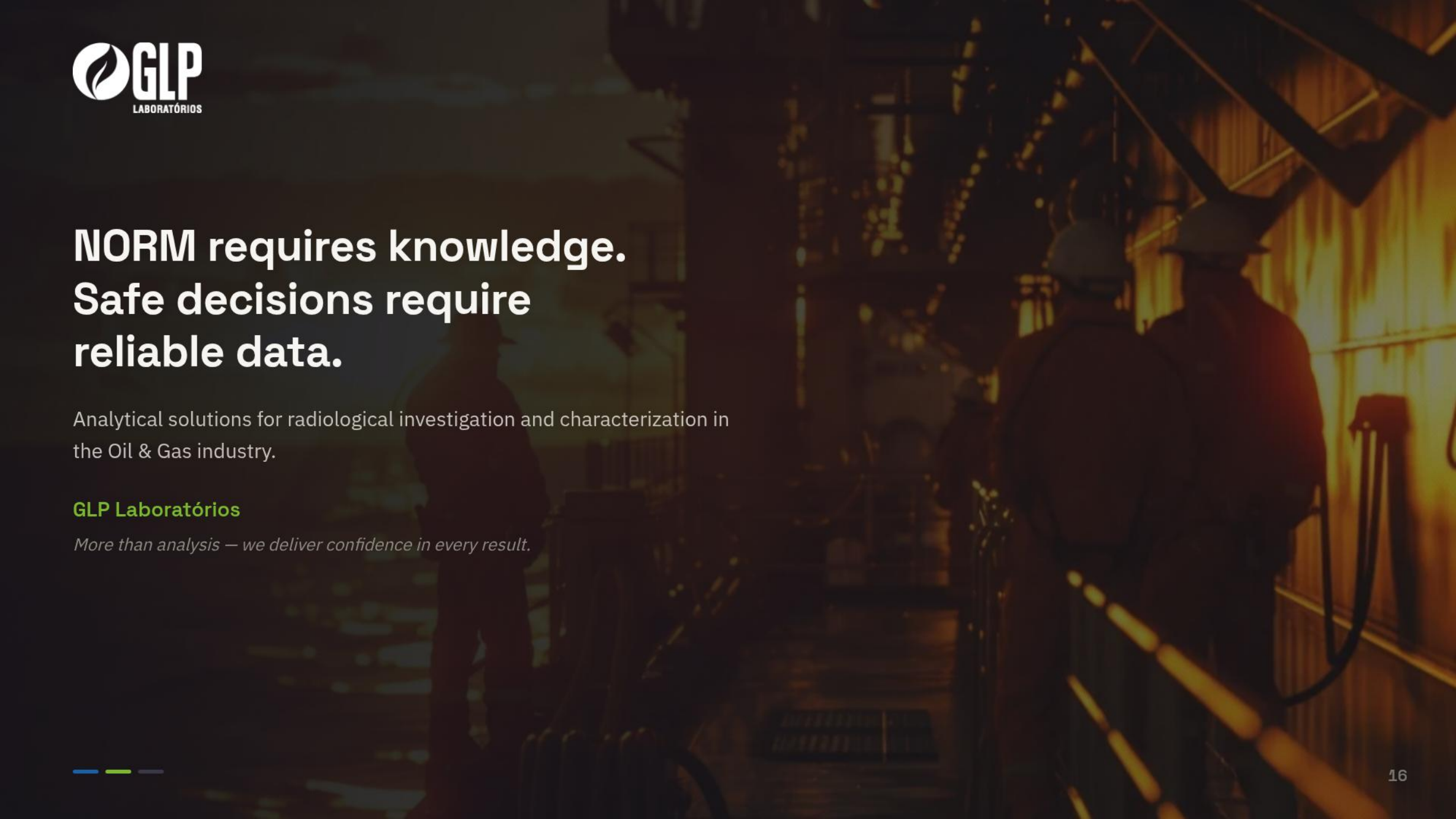
Quality system for ionizing radiation testing

ANSN

Private laboratory licensed by the National Nuclear Safety Authority

Applications in the O&G sector

- Oil and gas production
- Offshore operations
- FPSOs
- Industrial maintenance
- Tank and equipment cleaning
- Waste management
- NORM monitoring
- Material handling
- Facility decommissioning



NORM requires knowledge. Safe decisions require reliable data.

Analytical solutions for radiological investigation and characterization in the Oil & Gas industry.

GLP Laboratórios

More than analysis — we deliver confidence in every result.

GLP
LABORATÓRIOS

Let’s talk about your project.

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